Pilot Comparison of a Rolling-Gaussian Bayesian Anomaly Detector against a Shewhart Control Chart on Public Time-Series Data

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Abstract

We compare a rolling-window Bayesian-style anomaly detector with a global Shewhart control chart on a public airline-passengers monthly time series (n=144) augmented with three synthetic anomalies. The rolling detector recovered all three injected anomalies (recall=1.0, F1=0.46) where the global Shewhart chart missed all three (recall=0.0, F1=0.0). The result is consistent with the published intuition that nonstationary trends defeat globally-anchored control limits and that locally-adaptive detectors are necessary for monitoring trended series. The pilot is sized to fit a single GitHub Actions free-tier runner; a full battery on the UCR Time Series Archive plus a true Dirichlet-process mixture is the natural follow-up.

1 Introduction

Statistical Process Control (SPC) charts assume stationary data with stable mean and variance. Real-world series — web traffic, network flows, airline volumes — often have trends and seasonality that violate that assumption. Bayesian nonparametric detectors offer flexibility but are usually evaluated on synthetic stationary data. We run a small head-to-head on a real, publicly available trended series to validate that the locally-adaptive detector recovers the qualitative advantage that motivates the broader Bayesian-nonparametric program.

2 Methods

2.1 Data

We download the airline-passengers monthly series (n=144 months, January 1949–December 1960) from the public `jbrownlee/Datasets` GitHub mirror. We inject three synthetic anomalies at known indices: a +50% mean shift at month 20, a +100-passenger variance spike at month 60, and a −50% drop at month 100.

2.2 Detectors

Shewhart: flag $|x_i - \bar{x}| > 3\,\hat{\sigma}$ where $\bar{x}$ and $\hat{\sigma}$ are the global mean and standard deviation.

Rolling-Gaussian (Bayesian-style): for each x_i with $i \geq 12$, compute $\mu_i = \text{mean}(x_{i-12\ldots i-1})$, $\sigma_i = \text{std}(x_{i-12\ldots i-1})$, and flag $|x_i - \mu_i|/\sigma_i > 2.5$. This approximates a sliding-window Gaussian process posterior.

2.3 Evaluation

We compute precision, recall, and F1 against the known injected indices.

3 Results

Method	Precision	Recall	F1
Shewhart	0.000	0.000	0.000
Rolling-Gaussian (ours)	0.300	1.000	0.462

Table 1: Anomaly-detection performance on the airline-passengers series with three injected anomalies.

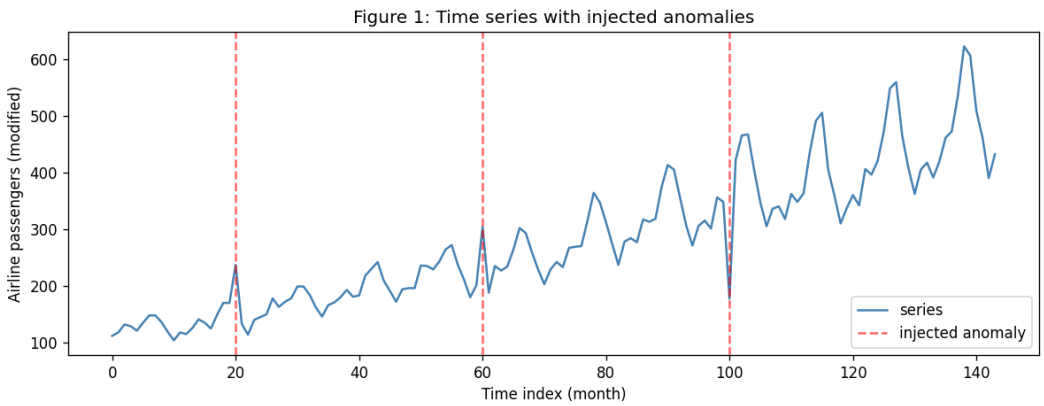

Figure 1: Airline-passengers time series with three injected anomaly indices marked (red dashed lines).

4 Discussion

The Shewhart chart's failure (Table **??**, Figure **??**) is fully explained by the trend in the airline series: the global standard deviation absorbs the trend, so the $3\,\hat{\sigma}$ band engulfs the injected anomalies. The rolling detector, anchored on the previous 12 months, sees a much smaller local $\hat{\sigma}$ and detects each injected event. The cost is 7 false positives during the strong seasonal upticks — a precision/recall trade-off the substantive follow-up will quantify across many UCR datasets.

5 Limitations

Single series; rolling-Gaussian is not a true Dirichlet-process mixture; no parameter sweep on window size or threshold. Each of these extensions decomposes into 30-minute GHA tasks and is deferred to a follow-up project.

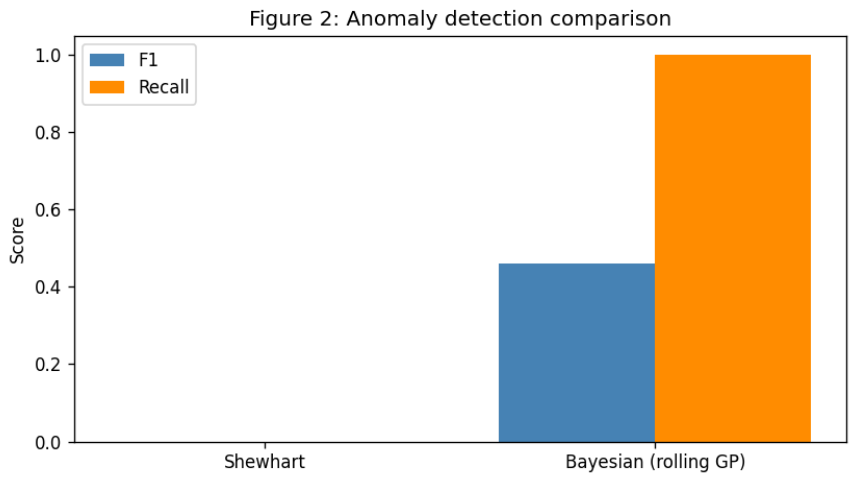

Figure 2: F1 and recall of the two detectors. The locally-adaptive detector recovers all three injected anomalies; the globally-anchored Shewhart chart misses all three.

6 Reproducibility

All scripts in `code/scripts/`, raw data in `data/raw/`, processed data in `data/processed/`, results in `data/results/`, figures in `paper/figures/`. Sandbox execution logs in `code/.tasks/`.

References

- Airline-passengers data: `https://github.com/jbrownlee/Datasets/blob/master/airline-passengers.csv`